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Unit – 4 : Session – 8 : SLO – 2 | AVL Tree — Insertion and Construction

■ I. Insert Sequence: 16, 27, 9, 11, 36, 54, 81, 63, 72

- Insert 16: tree = 16.
- Insert 27: right of 16. Insert 9: left of 16. Balanced.
- Insert 11: right of 9. Balanced.
- Insert 36: right of 27. Balanced.
- Insert 54: right of 36 -> node 27 unbalanced (BF -2, RR case) -> LEFT rotation at 27:
- 36 becomes subtree root: 16 -> left 9(right 11), right 36(27, 54).
- Insert 81: right of 54. Balanced.
- Insert 63: left of 81 -> node 54 unbalanced (RL case) -> RIGHT rotation at 81, then LEFT rotation at 54:
- 63 becomes subtree root: 36(27, 63(54, 81)).
- Insert 72: left of 81 -> node 36 unbalanced (RL case) -> RIGHT rotation at 81, then LEFT rotation at 36.
- FINAL AVL TREE:
- 16
- /\
- 9 63
- /\
- 11 36 81
- /\
- 27 54 72

■ II. Insert Sequence: 1, 2, 3, 4, 5, 6, 7, 8

- Insert 1, 2, 3 -> node 1 unbalanced (RR) -> LEFT rotation at 1: tree = 2(1, 3).
- Insert 4: right of 3. Balanced: 2(1, 3(-, 4)).
- Insert 5: right of 4 -> node 3 unbalanced (RR) -> LEFT rotation at 3: tree = 2(1, 4(3, 5)).
- Insert 6: right of 5 -> node 2 unbalanced (RR) -> LEFT rotation at 2: tree = 4(2(1, 3), 5(-, 6)).
- Insert 7: right of 6 -> node 5 unbalanced (RR) -> LEFT rotation at 5: tree = 4(2(1,3), 6(5,7)).
- Insert 8: right of 7 -> node 6 unbalanced (RR) -> LEFT rotation at 6:
- 7 becomes subtree root: 7(6(5, -), 8).
- FINAL AVL TREE:
- 4
- /\
- 2 7
- /\
- 1 3 6 8
- /
- 5
- Final: 4 -> left 2(1, 3); right 7(6(5, -), 8). All balance factors in {-1, 0, 1}.